

Solutions to Class Assignment on Buck Compensation

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Problem: Compensator Design with Type-III Compensation

a.

$$\frac{v_{out}}{V_{sw}} = \frac{1 + sR_{ESR}C_1}{s^2LC + s\frac{L}{R_{load}} + 1}$$

b. Modulator gain = $D/V_{\text{sawpeak}}=1$

c. Duty-cycle-output transfer function is given by

$$\frac{v_{out}}{\tilde{d}} = V_{in} \cdot \frac{1 + sR_{ESR}C_1}{s^2LC + s\frac{L}{R_{load}} + 1}$$

The following table lists the transfer functions.

Modulator gain	$\frac{D_{max}}{V_{saw_{pp}}}$
RLC transfer function	$\frac{1 + sR_{ESR}C_1}{s^2LC + s\frac{L}{R_{load}} + 1}$
Duty-to-output	$V_{in} \cdot \frac{1 + sR_{ESR}C_1}{s^2LC + s\frac{L}{R_{load}} + 1}$
Control (Vc) to output	$\frac{D_{max}}{V_{sawpp}} \cdot V_{in} \cdot \frac{1 + sR_{ESR}C_1}{s^2LC + s\frac{L}{R_{load}} + 1}$

d. We use a type 3 compensator as shown in the following figure.

1. Notice that the mid-band gain of the error amplifier is multiplied with the power path gain (= 2.5 or ~ 8 dB) in this problem. Therefore, in order to achieve a high phase margin of greater than 45°, the mid-band gain of the compensator is designed to be 10 dB so that the overall mid-band gain is close to 20 dB.
2. You can push the error amplifier UGB or Gain Bandwidth to a high frequency in order to make sure that it does not add a pole to the loop and make it unstable. I have pushed it to 100 Meg.
3. The first zero of the loop given by $1/(R_2C_2)$ is set to be 1/10 of ω_o to provide adequate phase boost before the output pole pair.
4. The second zero, given by $1/(C_3(R_1+R_3))$ is set to be at ω_o .
5. The pole given by $1/(C_3R_1)$ is set to be 2. ω_u for a phase margin ~60°. If it is set at ω_u , the phase margin must be ~45°

Using $f_o = 1/(2\pi\sqrt{LC})=63.6$ kHz,

$$\frac{1}{R_2 C_2} = \frac{63.6 \cdot 10^3 \cdot 2 \cdot \pi}{10} \quad (1)$$

$$20 \cdot \log \frac{R_2}{R_1} = 10$$

$$\frac{R_2}{R_1} = 3.16$$

Set $R_1=10k$,

and $R_2=31k$

$C_2=1$ nF from (1).

Also,

$$\frac{1}{(R_3 + R_1)C_3} = 63.6 \cdot 10^3 \cdot 2 \cdot \pi \quad (2)$$

For a PM of 45° ,

$$\frac{1}{R_1 C_3} = 10 \cdot 63.6 \cdot 10^3 \cdot 2 \cdot \pi \quad (3)$$

From (2) and (3),

$$R_1 \sim \frac{R_3}{10} = 1 k$$

Solving (3), $C_3 = 250$ pF



